WaveFactor enables factor inference for spatial transcriptomics with multi-scale spatial awareness and prior information integration

Yulin Wu[1, 2], Heejung Shim[1, 2, 3, 4]

[1] Melbourne Integrative Genomics & School of Mathematics and Statistics, The University of Melbourne
[2] ARC Centre of Excellence for the Mathematical Analysis of Cellular Systems, The University of Melbourne
[3] Snow Vision Accelerator & Charles Perkins Centre, The University of Sydney
[4] Sydney Precision Data Science Centre & School of Mathematics and Statistics, The University of Sydney

Introduction

Spatial transcriptomics enables studying gene expression with spatial context. Challenges in analysing spatial transcriptomics include high dimensionality and complex multi-scale spatial structures. Confronted with this, one possible direction is to leverage existing prior information, such as marker genes, pathway genes, and expected scale of significant signals.

Motivated by this, we developed WaveFactor, a wavelet-based Bayesian sparse factor model for spatial transcriptomics accounting for multi-scale spatial and prior information.

WaveFactor learns latent factors whose activity scores constitute a low-dimensional representation of the input spatial transcriptomics data usable for further downstream analysis. When inferring factors, WaveFactor leverages wavelet transform to capture multi-scale spatial information and accepts prior information with adjustable strength to be accounted for probabilistically. Additionally, WaveFactor outputs scores enabling quantitative comparison of factor importance and gene importance for factors.

WaveFactor is available for use as a Python package.

Method

Input	Spatial transcriptomics data (e.g. Visium, Stereo-seq) (Optional) Prior gene sets, multiscale signal vs noise presence

1. Apply wavelet transform → wavelet-gene expressions Y (multi-scale spatial data representation of gene expression)
2. Fit factor model with variational inference → factor-wavelet activities L, factor-gene loadings F
3. Apply inverse wavelet transform to factor-wavelet activities L → factor-spot activities $\mathcal{L}$

Output	Estimates for factor-gene loadings F, factor-wavelet activities L, factor-spot activities $\mathcal{L}$

Model

Variables	
Y	Spatial transcriptomics data (after wavelet transform)
Z_L	Factor-wavelet relation (Yes (1) or No (0))
L	Factor-wavelet activities: strength of factor-wavelet relation
Z_F	Factor-gene relation (Yes (1) or No (0))
F	Factor-gene loadings: strength of factor-gene relation

Equations	
$Y\|L,F,\tau \sim N(LF,\tau^{-1})$	
$L\|Z_L,t \sim \begin{cases} 0, Z_L = 0 \\ N(0,t^{-1}), Z_L = 1 \end{cases}$	$F\|Z_F \sim \begin{cases} 0, Z_F = 0 \\ N(0,1), Z_F = 1 \end{cases}$
$Z_L \sim \text{Bernoulli}(\pi_L)$	$Z_F \sim \text{Bernoulli}(\pi_F)$
$t \sim \text{Gamma}(\alpha_t, \beta_t)$	$\tau \sim \text{Gamma}(\alpha_\tau, \beta_\tau)$

Note: π_F, π_L encode prior gene sets, multiscale signal vs noise presence

Application: Align and Refine Pathways

We applied *WaveFactor* on MOSTA [1] mouse embryo data with partial pathway gene sets (leave out 10%) as priors. We found *WaveFactor*:

- Learnt factor activities capturing expected pathway activities
- Aligned factors across samples without needing post-hoc matching
- Refined prior pathway gene set by recovering left out genes from data

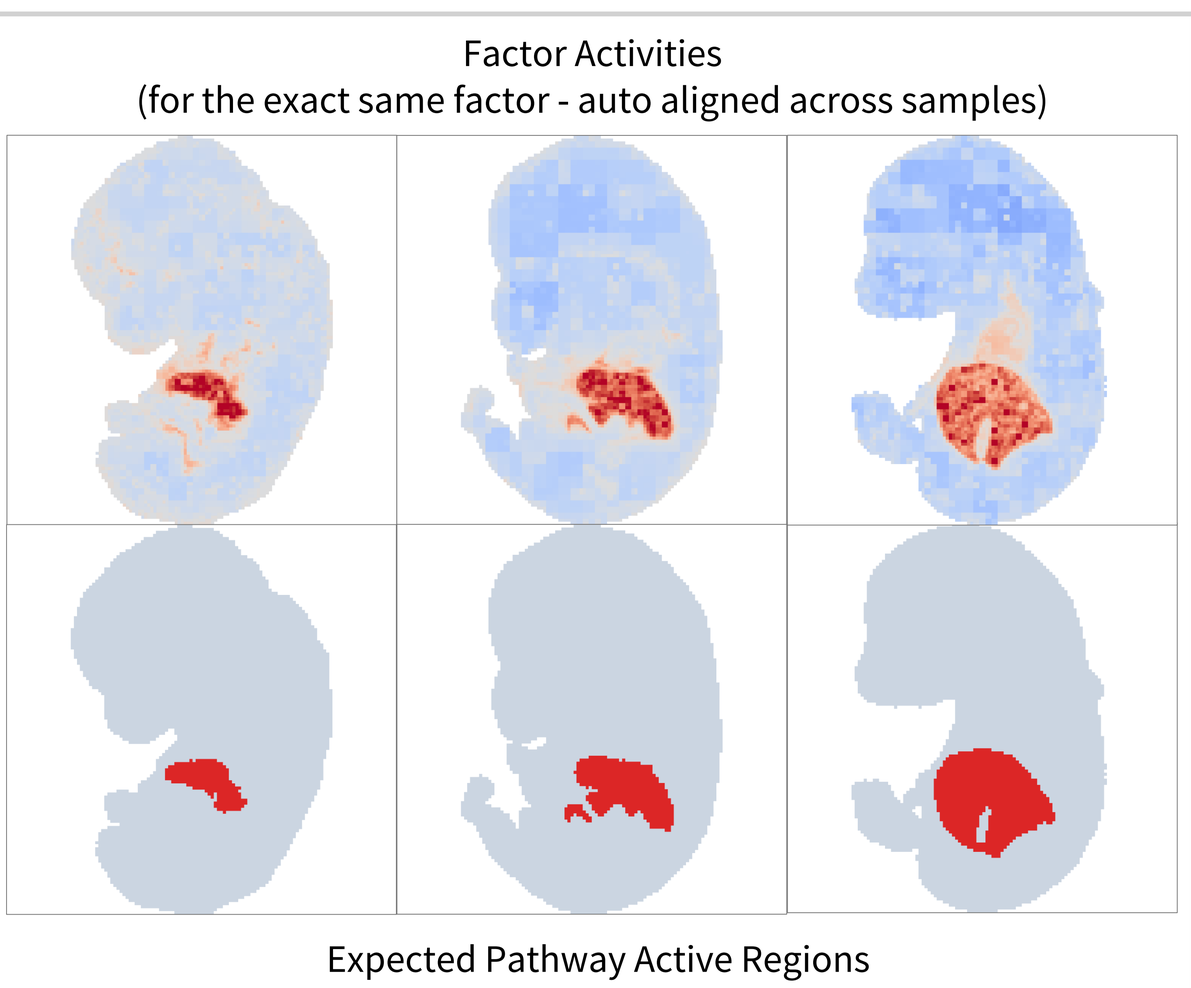


FIGURE: For selected factor given heme biosynthetic process pathway (GO:0006783) genes as prior, factor activities matched (liver) annotations where this pathway was expected to be active in such embryonic samples. Columns show activities & annotations for each sample.

Performance

WaveFactor runtime scales linearly with the number of spots, genes, factors, assuming that the *WaveFactor* model fitting algorithm converges in a constant number of iterations.

In practice, the number of iterations to convergence varies depending on the data and *WaveFactor* settings such as convergence tolerance.

Dataset Name	Platform	Data Dimensions	Runtime
Mouse Embryo Sample E11.5 E1S1 [1]	Stereo-seq	$N = 16384$ $G = 3000$	17.3s
Human DLPFC Sample 151673 [2]	Visium	$N = 4096$ $G = 3000$	6.7s

TABLE: Illustrative *WaveFactor* runtimes on single CPU core. N, G are respectively the number of spots, genes of the preprocessed data inputted into *WaveFactor*. Further settings omitted for brevity.

Conclusion

WaveFactor is a wavelet-based Bayesian sparse factor model for spatial transcriptomics accounting for multi-scale spatial and prior information.

WaveFactor is available for use as a Python package.

WaveFactor

shimlab/WaveFactor

https://shimlab.github.io/WaveFactor/

References

[1] Chen et al, *Cell*, 2022
[2] Maynard et al., *Nature Neuroscience*, 2021